PARTH SANGHAI

Electrical & Electronics Engineering | BITS Pilani, Goa

CONTACT

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TECHNICAL SKILLS

- Computer Vision (YOLOv11, OpenCV)
- Verilog / Digital Design
- 8086 Assembly Language
- Signal Processing
- Python / Machine Learning
- FastAPI / REST APIs
- Circuit Analysis
- Git / Version Control

CERTIFICATIONS

Spacecraft & Satellite Engineering
Udemy — Interplanetary Systems

Machine Learning — Andrew Ng
Coursera / DeepLearning.AI

INTERESTS

- Deep Space Astrophysics
- Competitive Football
- Hardware Prototyping

PROFILE

Second-year B.Tech student in Electrical & Electronics Engineering at BITS Pilani, Goa, with hands-on experience across computer vision, hardware design, signal processing, and low-level software. Completed a practical training placement at BISAG-N (MeitY, Govt. of India) building an AI-based PCB inspection system. Curious by nature — from processor-level game logic to spacecraft orbital dynamics.

EDUCATION

B.Tech — Electrical & Electronics Engineering | *BITS Pilani, Goa (2024 – Present)*

- Currently in 2nd year of the B.Tech programme.

TECHNICAL PROJECTS

Computer Vision-Based PCB Component Scanner & BOM Generator | *PS-I Practical Training, BISAG-N (MeitY, Govt. of India)*

Bhaskaracharya National Institute for Space Applications & Geo-informatics, Gandhinagar | May – July 2026

- Built NeuralBoard, a full-stack web app that scans PCB/motherboard photos and auto-generates a Bill of Materials via a dual YOLOv11 pipeline (detector + classifier).
- Trained a YOLOv11 detector across iterative runs, raising overall mAP50 from 0.162 to 0.460 by integrating custom-annotated boards with the PCB WACV 2019 and SMD4K2 datasets; diode and transistor mAP50 rose from 0.12/0.04 to 0.93/0.77.
- Trained a companion YOLOv11 classifier to 99.49% Top-1 accuracy across 37,023 component crop images spanning 6 classes.
- Built a FastAPI backend serving YOLO11 inference and a JS/HTML5 frontend with live camera capture, canvas bounding-box overlays, and CSV/JSON/PDF export.

Tech stack: Python, Ultralytics YOLOv11, PyTorch (CUDA), OpenCV, FastAPI, JavaScript, jsPDF, Roboflow, Git.

8086 Whack-a-Mole | *Microprocessors & Interfacing*

- Developed a fully interactive game in 8086 Assembly Language from scratch, handling real-time I/O, timing control, and hardware interfacing at processor level.
- Demonstrates mastery of interrupt handling, port I/O, and low-level memory management.

Seismic Noise Filtering System | *Signals & Systems*

- Designed a signal processing pipeline using frequency-domain analysis to filter noise from real seismic wave data and improve signal clarity.
- Presented findings via a collaborative live demo.

ACHIEVEMENTS & ACTIVITIES

Football — Semi-Professional | *Mumbai & Maharashtra*

- Represented Mumbai at state-level competitions; shortlisted for Maharashtra national squad.
- Played in the Youth I-League for Community Football Club of India — top-tier youth competition.

Event Coordination & Volunteering

- Volunteered at International Rover Challenge — contributed to large-scale event operations.
- Served as official Referee for a veteran table tennis tournament, ensuring fair and smooth play.